

Module 14: Macroevolution — Questions

Introduction

The following questions are designed to test your knowledge of macroevolution, the development of new species (speciation), and the reproductive barriers that keep biological groups genetically distinct. Use this simple list to verify your understanding of how macro-level changes emerge over long timeframes.

Practice Questions

1. How does the Biological Species Concept define a species?
2. What is a reproductive barrier, and what is its role in keeping closely related species separate?
3. What is the fundamental difference between prezygotic and postzygotic reproductive barriers?
4. Give an example of a habitat (ecological) prezygotic isolation mechanism.
5. Provide a real-world example of temporal isolation.
6. How does behavioral isolation prevent reproduction between two groups?
7. Describe mechanical isolation and explain why it stops mating before fertilization can occur.
8. What is gametic isolation and in what types of organisms is it most commonly observed?
9. Explain the concept of reduced hybrid viability (a postzygotic barrier).
10. What occurs in the postzygotic barrier known as reduced hybrid fertility?
11. Describe hybrid breakdown and how it impacts successive generations.
12. What is allopatric speciation and what usually causes it?
13. How does sympatric speciation occur without any geographic separation?
14. How can polyploidy (having extra sets of chromosomes) lead to instant sympatric speciation in plants?
15. In what way are the small, generation-to-generation changes of microevolution connected to the massive species-level shifts seen in macroevolution?